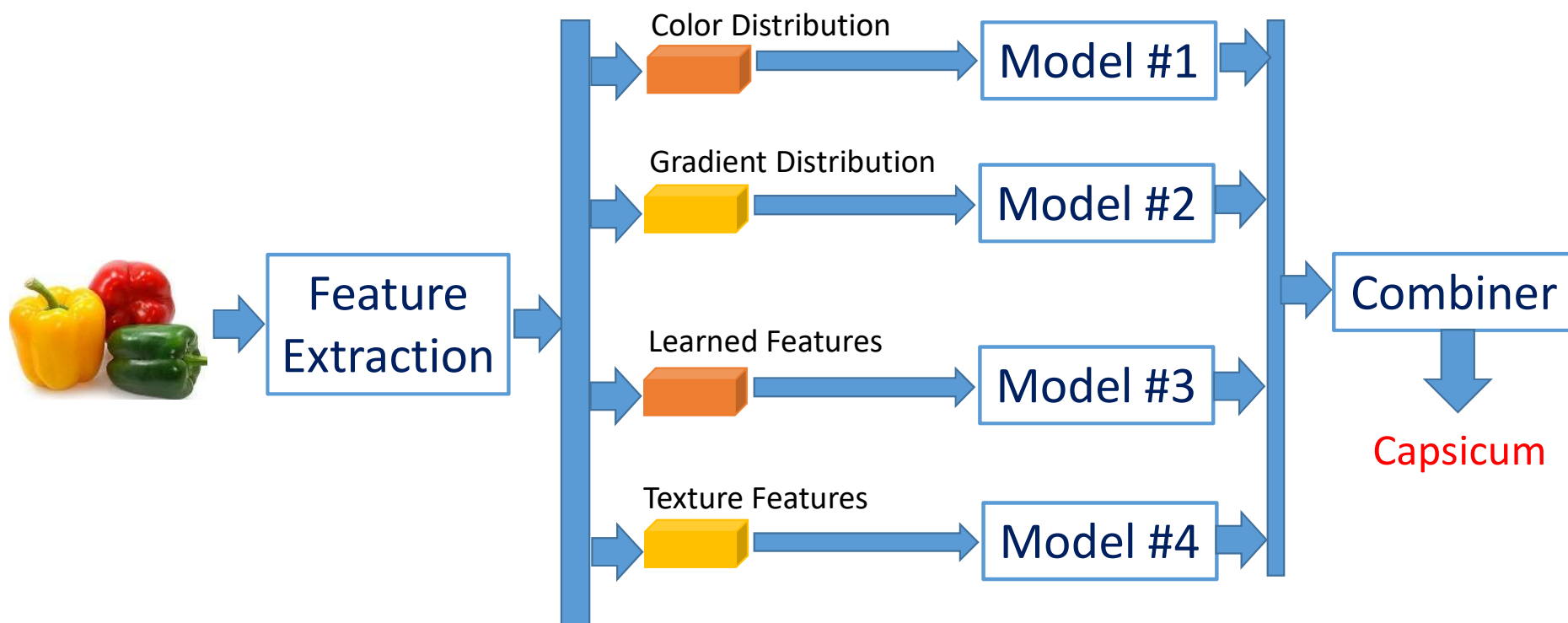


Decision Forest



Prithwijit Guha
Dept. of EEE, IIT Guwahati

Late Fusion in Ensemble Framework

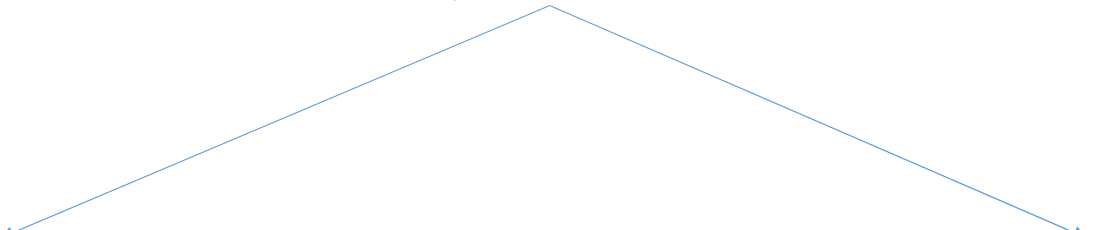


Late Fusion: Ensemble Techniques

- Diversity Induction in Training Dataset
- Weak Learners Trained on Diverse Datasets
- Weak Learners Provide Intermediate Predictions
- Multiple Predictions Fused at Combiner Stage
- Ensemble using Bagging and Boosting
- Boosting – Cascade of Weak Learners
- Bagging – Bootstrap Aggregating
- Weak Learners Trained on Diverse Datasets

Random Forest: Diversity Induction

$$\mathcal{S} = \{(\mathbf{x}_i, \mathbf{y}_i); i = 1, \dots, n\}$$

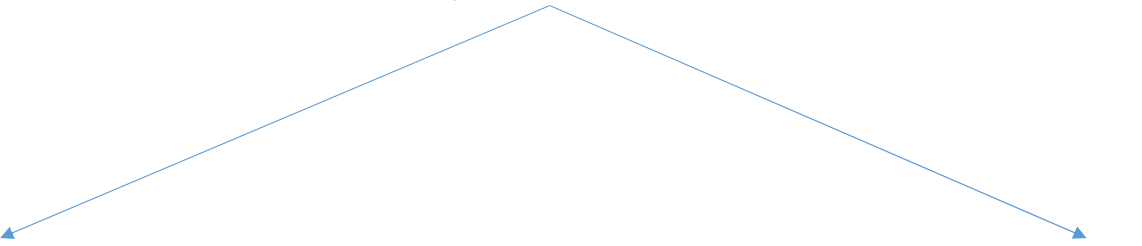

$$\mathcal{S}_1 = \{(\mathbf{x}_i^{(1)}, \mathbf{y}_i^{(1)}); i = 1, \dots, n_1\} \cdots \mathcal{S}_m = \{(\mathbf{x}_i^{(m)}, \mathbf{y}_i^{(m)}); i = 1, \dots, n_m\}$$

$$\mathcal{S} = \bigcup_{r=1}^m \mathcal{S}_r$$

$$n_r = \alpha n$$

$$\alpha \in (0,1)$$

Random Forest: Diversity Induction

$$\mathcal{S} = \{(\mathbf{x}_i, \mathbf{y}_i); i = 1, \dots, n\}$$


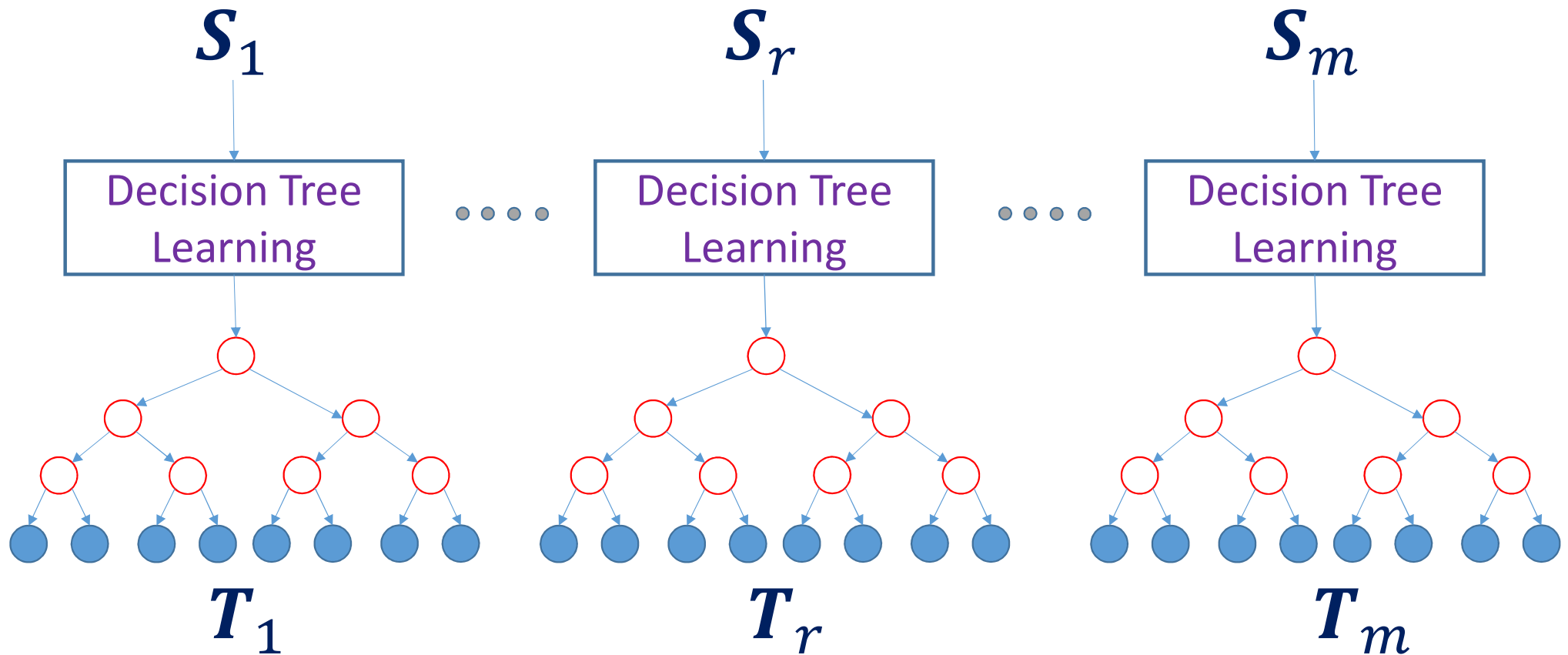
$$\mathcal{S}_1 = \{(\mathbf{x}_i^{(1)}, \mathbf{y}_i^{(1)}); i = 1, \dots, n\} \cdots \mathcal{S}_m = \{(\mathbf{x}_i^{(m)}, \mathbf{y}_i^{(m)}); i = 1, \dots, n\}$$

$$\mathbf{x}_i^{(r)} = M_r \mathbf{x}_i \quad \left\{ \begin{array}{l} \mathbf{x}_i \in \mathbb{R}^d \\ \mathbf{x}_i^{(r)} \in \mathbb{R}^q \\ M_r \in \{0,1\}^{q \times d} \end{array} \right. \quad q < d$$

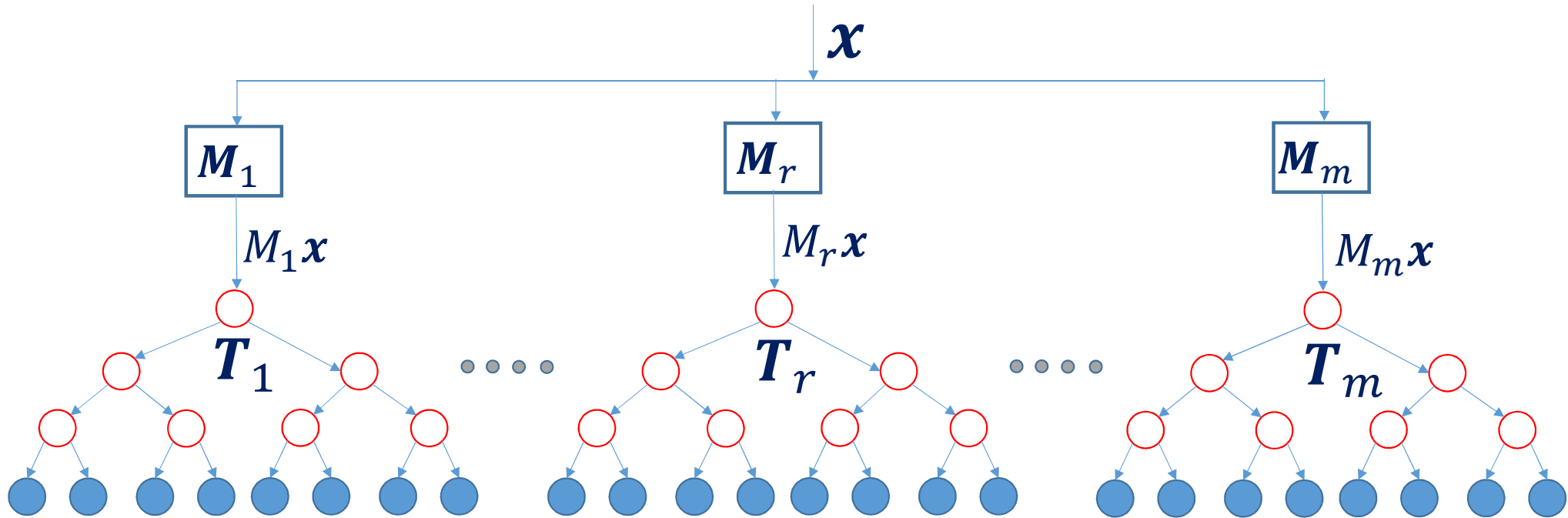
Example

$$\begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Random Forest: Learning



Random Forest: Operation



$$\hat{y} = \frac{1}{m} \sum_{r=1}^m T_r(M_r x)$$

Rotation Forest: Diversity Induction

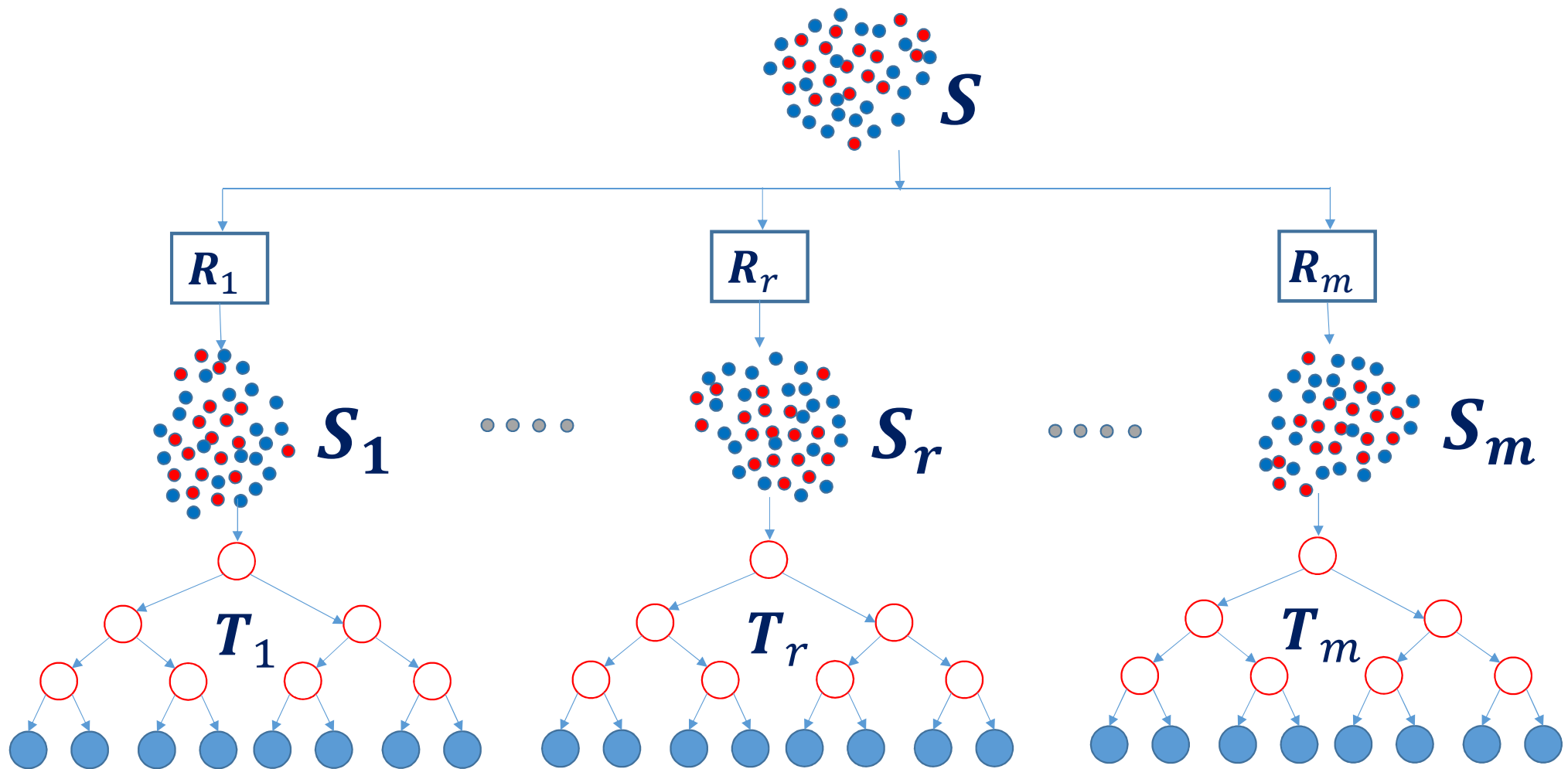
$$\mathcal{S} = \{(\mathbf{x}_i, \mathbf{y}_i); i = 1, \dots, n\}$$

$$\mathcal{S}_1 = \{(\mathbf{x}_i^{(1)}, \mathbf{y}_i^{(1)}); i = 1, \dots, n\} \cdots \mathcal{S}_m = \{(\mathbf{x}_i^{(m)}, \mathbf{y}_i^{(m)}); i = 1, \dots, n\}$$

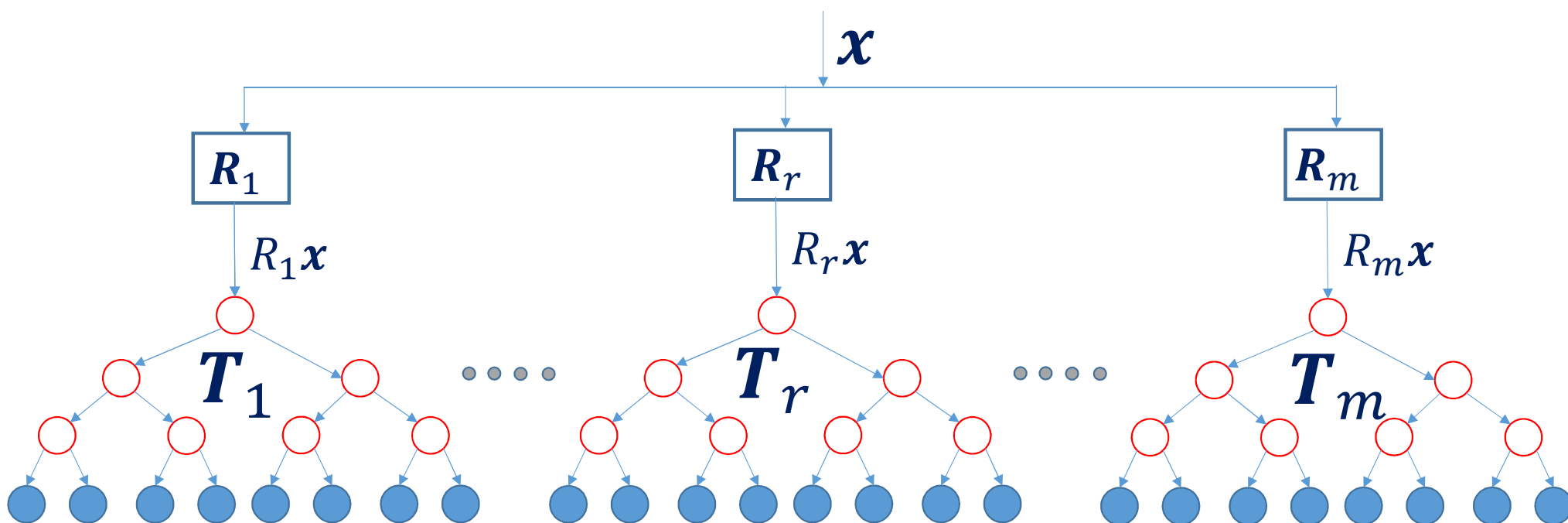
$$\mathbf{x}_i^{(r)} = R_r \mathbf{x}_i \left\{ \begin{array}{l} \mathbf{x}_i \in \mathbb{R}^d \\ \mathbf{x}_i^{(r)} \in \mathbb{R}^d \\ R_r \in SO(d) \end{array} \right.$$

- Generate Random $d \times d$ Matrix A
- Create Matrix $B = A + A^T$
- Compute Eigen Vectors $\hat{\mathbf{e}}_i$ ($i = 1, \dots, d$) of B
- Construct $R = [\hat{\mathbf{e}}_1 \quad \cdots \quad \hat{\mathbf{e}}_d]$

Rotation Forest: Learning



Rotation Forest: Operation



$$\hat{y} = \frac{1}{m} \sum_{r=1}^m T_r(R_r \mathbf{x})$$

Summary

- Ensemble Methods – Bagging & Boosting
- Decision Forest – Collection of Trees
- Random Forest – Diversity Induction in Feature Subspaces
- Rotation Forest – Diversity Induction by Random Rotations



Thank You